# PipelineFortress

Secure CI/CD with GenAI-assisted triage. CAP-DSE-3W individual study implementation and evidence report.

| Six security stages | Application tests | Underlying issues |
| --- | --- | --- |
| Executed locally | 2332 passed / 6 skipped | 180 open and deferred |

## Outcome and operating decision

The owned lab now executes SAST, dependency, secret, IaC, image and passive runtime scans. A hardened source-built image is linked through deployment, container inventory and scan evidence. Source-free normalization, mock triage, refusal tests and bounded local patch templates are implemented.

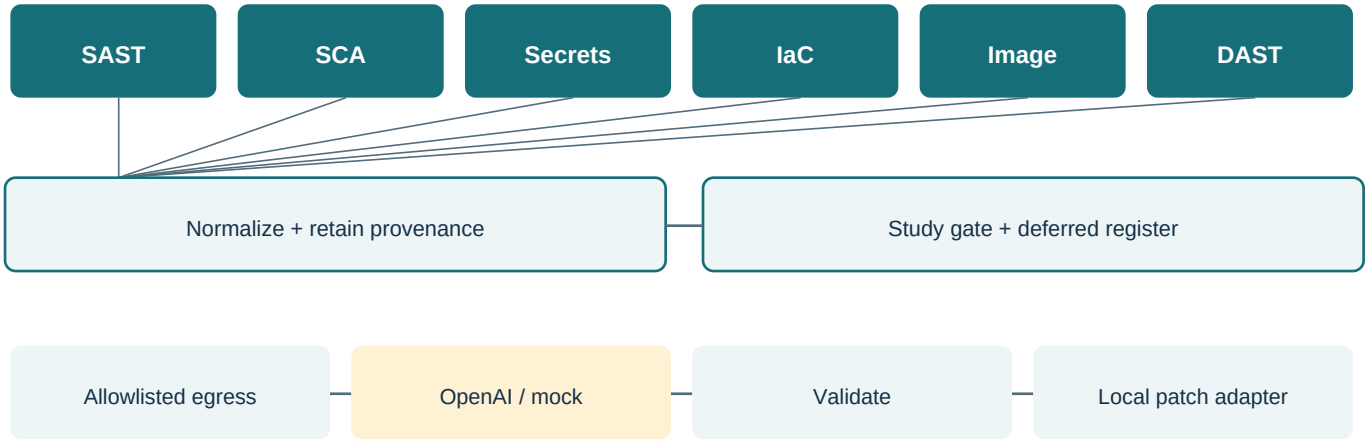
The project owner subsequently requested fixing all critical findings. Current source and image scans report zero critical findings after dependency replacement and authentication/query hardening. A new zero-critical gate rejects regressions; lower severities remain visible and deferred. Production readiness is not claimed.

## Completion boundary

Local pipeline execution and the three example proposal regressions are verified. Successful live API triage remains unverified: the selected OpenAI credential passed model lookup but three bounded generation attempts returned HTTP429. Genuine human review is recorded only when received. Hosted and fresh-directory status are shown explicitly later in this report.

This report is an evidence snapshot, not an assertion that every assessment dependency has been fulfilled.

# Architecture and trust boundaries




Every scanner retains raw reports in ignored local storage and publishes only execution receipts, aggregate counts, opaque finding fingerprints and checksums. Job receipts bind one run and commit; build, container and DAST receipts must agree on the image configuration ID.

Normalization creates observation IDs and underlying-issue IDs. Source/container dependency observations share an issue when advisory, component and version agree; detecting stages and every source record remain available. A matching title alone is not a deduplication key.

The AI boundary constructs a new allowlisted object. It sends synthetic IDs, verified rule/advisory references, severity/category, aliases and verified public versions. Private locations, source, matches, account details and raw scanner messages remain local. The model receives no shell, repository, file or write tools.

Validated operations map to fixed local templates. A human review step is separate from generation, scanner success and author self-review. No automatic merge is implemented.

# Environment and application verification

| Component | Verified configuration |
|---|---|
| Target | OWASP Juice Shop 20.3.0; pinned upstream base with documented critical-remediation modifications; original license retained |
| Node / Python | Build Node 24.22.0; runtime 24.22.1; isolated Python 3.10 |
| Scanners | Semgrep 1.181.0; Trivy 0.76.0; Gitleaks 8.30.1 |
| IaC / inventory / runtime | Checkov 3.3.30; Syft 1.55.0; ZAP 2.17.0 |
| Kubernetes lab | kind 0.33.0; Kubernetes/kubectl 1.37.0 |
| Installation boundary | Project/user-local tools; scanner distributions and base images pinned |

The fresh offline application profile passes root/frontend installation, explicit production build, frontend tests, server tests and local API tests. Counts are 1372 frontend, 422 server and 538 API tests passed, with 6 upstream skips and zero selected failures.

A separate online attempt retained six failures from external-service DNS lookup errors. The offline profile excludes the entire nine-test internet-resource availability suite. It does not suppress a local application test failure. Historical online baseline counts remain separate.

The historical upstream import remains traceable. The current remediation variant updates the root dependency lock, replaces unsafe dependencies and hardens JWT and database queries; the frontend lock is preserved. Runtime input hashes and local project-tree hashes bind the changed source and exact image. These local changes have not yet been verified in hosted CI.

# Observed runtime baseline

| Observation | Hardened runtime result | Interpretation |
|---|---|---|
| Home / products / languages | HTTP200; data present | Application remains usable |
| Coding snippet | HTTP200 | Training source snippets remain available locally |
| FTP directory | HTTP200 | Intentional directory listing remains |
| Metrics | HTTP200 | Unauthenticated metrics remain observable |
| Malformed product search | HTTP500 with SQL error shape | Training error disclosure remains |

Seven functional/training checks pass in Docker and Kubernetes. The observations are against the owned lab only. No external exploitation, authenticated coverage or customer-data access is claimed.

Infrastructure hardening preserves the intentionally vulnerable application. Its behavior is part of the study baseline and remains in the deferred register. A passing runtime check establishes the expected behavior, not absence of security weaknesses.

The active hardened workload is Ready with zero restarts at verification. Earlier readiness issues in other pre-existing system pods are retained as a lab limitation; application readiness is not a complete cluster-health or CIS assessment.

# Six stages and the authoritative verdict

| Stage | Normalized observations | Execution |
|---|---|---|
| SAST | 11 | Complete; findings deferred |
| SCA | 23 | Complete; findings deferred |
| SECRETS | 133 | Complete; findings deferred |
| IAC | 1 | Complete; findings deferred |
| CONTAINER | 41 | Complete; findings deferred |
| DAST | 42 | Complete; findings deferred |

The complete local run is 20261022T181348556832Z. Its aggregate verdict is PASS_WITH_DEFERRED_RISKS, with 0 operational errors. The stage receipts retain raw report hashes and exact image linkage.

That execution took 93.19 seconds after explicit reuse of a freshly verified build/test. The application check stages took 143.51 seconds separately. Bootstrap/download costs are additional; no percentile is inferred.

The hosted workflow defines six scan jobs, build/test and one final aggregate job. It uses read-only PR permissions, pinned action SHAs, full history for secrets, an integrity-checked image archive between jobs and cleanup of its own staging cluster. Skipped jobs, malformed/missing receipts, scanner errors and mixed identities cannot pass the execution check.

Earlier hosted baseline, before critical remediation: completed; conclusion: success. Run: https://github.com/swetha12g/IITR-Capstone-Proj/actions/runs/39821125926

# Hardening and software inventory

| Measure | Baseline | Hardened |
|---|---|---|
| Failed Kubernetes checks | 15 | 1 |
| Failed Dockerfile checks | 2 | 0 |
| Image vulnerability identities | 101 | 100 |
| Critical / high / unknown | 11 / 27 / 1 | 11 / 27 / 0 |
| Image bytes | 351,789,266 | 345,391,667 |
| CycloneDX components | Not claimed here | 2748 |

The runtime uses UID/GID 65532, no effective or bounding capabilities, no-new-privileges, RuntimeDefault seccomp and a read-only root filesystem. Application code and immutable seed reject writes. Explicit bounded writable mounts preserve SQLite, restored training files, translations, customized frontend media, uploads, logs, CSAF metadata and temporary files.

In the historical Day 4 comparison, both immutable image configuration IDs were scanned under one frozen Trivy database. One unknown CA-certificate advisory was removed; no new identity was introduced in that comparison. That comparison predates the later zero-critical remediation; current residual counts are shown separately. Syft CycloneDX 1.6 validates against hash-pinned official schemas and representative installed package versions.

Selected controls are mapped to CIS Kubernetes 2.0.0. NetworkPolicy remains a declared gap and enforcement is unverified. No full benchmark or cluster certification is claimed. A Docker configuration ID, a registry base index digest and a local containerd manifest digest are different identifiers.

# Passive DAST and coverage limits

| Item | Observed result |
| --- | --- |
| Scanner | ZAP 2.17.0; official image pinned by digest |
| URLs recorded | 121 |
| Alert groups by risk code | {"2": 2, "1": 8, "0": 4} |
| Verified seed routes | Home, product search, languages, FTP and metrics |
| Rule policy | Selected header rules retain FAIL classification; study verdict defers findings |
| Isolation / cleanup | Owned node network namespace; loopback API; scanner container removed |

The scan uses a traditional unauthenticated crawler and passive checks. It does not establish full single-page-app navigation, login/role coverage, active attack coverage, business-logic correctness or endpoint completeness. Passive alert-instance limits are retained as scanner coverage limits.

Early scanner-start attempts exposed loopback IPv6 and API Host-header allowlist mismatches. The corrections allow only the local API client and retain diagnostics. Successful scanning is supported by recorded app routes and report hashes, rather than a container exit code alone.

A different-UID container probe reproduces the hosted workdir write failure, then verifies the group-access repair. The scanner retains its own UID; only the owned mounted workdir receives group write access. Selected header failures are also detected before and absent after repair in isolated fixtures.

HTTP findings remain visible in normalization and the deferred register. The security result is not rewritten as clean merely because assignment execution continues.

# Detection evaluation and dashboard

| TP | FP | FN | TN | Labeled cases |
|---|---|---|---|---|
| 14 | 0 | 1 | 6 | 21 |

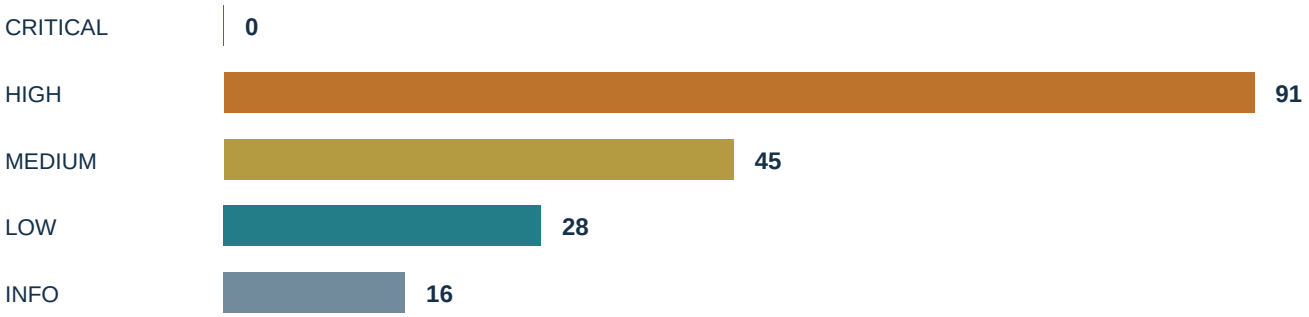
| Precision | Recall | F1 | Known introduced / caught |
|---|---|---|---|
| 100.0% | 93.3% | 96.6% | 15 / 14 |

The curated set combines deliberately unsafe and repaired source, advisory, canary, configuration, image and response-header cases. It includes a real scanner miss: the six authored SAST pattern rules do not detect an unprotected synthetic export route. That fixture is never executed against a real service.

The measured 93.3% fixture recall exceeds the scenario target on this small set. Labels were curated after the broader study work and are not independently human reviewed. The result is not an estimate of all real-world vulnerabilities, a blinded benchmark or evidence that remaining app defects were found.

The dashboard recomputes from normalized evidence and labels, exports JSON/CSV, separates observations from underlying issues and displays known misses. Empty denominators are N/A. It does not present percentiles from one timing observation. Current global underlying issue count is 180.

## Deferred underlying issues by severity



# AI safeguards and honest live status

| Boundary | Implemented evidence |
| --- | --- |
| Input construction | Exact field allowlist; synthetic IDs and aliases; independent scanner-derived version catalog |
| Refusal behavior | Unknown fields, invalid schemas, unverified/encoded metadata and duplicate IDs refuse before transport |
| Actual HTTP body | Mock captures final serialized request; private paths/source/canaries excluded |
| Grounding | Unknown IDs, invented CVEs/versions, unsupported operations and omissions rejected |
| Live restrictions | OpenAI only; no tools; store=false; bounded tokens, attempts and conservative reservations |

64 tests cover the relevant normalization, aggregate, boundary, grounding and bounded-patch behavior. A twenty-finding mock batch validates; mock output is deterministic policy behavior and is not reported as a real model evaluation.

The user-selected existing OpenAI key passed read-only model lookup. All three bounded generation attempts returned HTTP429; a specific underlying cause was not established. No successful model output or actual token/invoice cost is fabricated. The request cap is exhausted. No credits were purchased, account topped up or billing settings changed.

OpenAI documents standard abuse-monitoring retention separately from response storage. store=false does not establish zero retention or an account-specific ZDR setting. Only source-free metadata was submitted, and account-level retention settings were not independently verified.

# Three focused remediation examples

| Proposal | Control | Measured verification |
|---|---|---|
| pf-001 | disable-sa-token | Checkov fails before / passes after; both deployments Ready; home/products pass; namespace cleaned |
| pf-002 | require-nonroot | Checkov fails before / passes after; both deployments Ready; home/products pass; namespace cleaned |
| pf-003 | add-liveness-probe | Checkov fails before / passes after; both deployments Ready; home/products pass; namespace cleaned |

These examples modify only the legacy baseline deployment manifest through fixed local templates. They disable unnecessary service-account token mounting, explicitly require non-root execution and add a liveness probe. The original baseline image already runs non-root; the second proposal adds an explicit policy guarantee rather than claiming an observed root process was removed.

Each proposal has a bounded single-file diff, before/after hashes, actual selected-check evidence, functional observations and rollback instructions. Unsupported operations, path traversal, credential access, arbitrary commands and workflow edits are outside the adapter allowlist.

The Codex coding assistant authored the templates. The unavailable external application API is not credited with generating them. Three examples do not close the larger risk backlog, which remains deferred by the user.

Earlier infrastructure draft PR links recorded: 3. Genuine review/first-review correctness is pending unless the review ledger records a real verdict. No author self-approval, fabricated independent review or automatic merge is claimed.

# Reproduction and maturity self-assessment

Reproduction bootstraps project-local Python/pip/Node and verified scanner/tool distributions, runs the selected application suites, rebuilds the image, scans all stages, deploys into a uniquely named temporary kind cluster, runs passive DAST, normalizes, checks mock AI safeguards and regenerates the dashboard. Live API usage is excluded from reproduction by default.

Fresh-directory result: PASS_WITH_DEFERRED_RISKS, commit 1e0115c. Temporary-cluster cleanup verified: True. This is a fresh project directory on the same Docker host with standard caches, not a fresh-machine test. The later isolated DAST fixture was verified separately. The original study cluster is a separate resource.

## OWASP SAMM v2 lens

| Practice | Baseline | Current lab evidence / limit |
|---|---|---|
| Secure Build | Manual setup and selected tests | Repeatable pinned build, dependency inventory and automation; mature production prevention not claimed |
| Security Testing | Runtime observations | Six scanner stages and negative controls; known misses and limited DAST coverage retained |
| Defect Management | Findings not consolidated | Deduplicated register, labels and focused proposals; real human reviews still required |
| Strategy and Metrics | Planning estimates | Recomputable counts, fixture precision/recall and execution timings; one lab/run is limited evidence |

SAMM uses maturity activities across three levels and fifteen practices. Selected level-one and automation activities are evidenced here, but this short project does not complete a formal SAMM assessment or justify an overall organizational score. Deferred production defects and missing independent review prevent a mature-production assurance claim.

# Ethics, residual risks and references

The target is the owned local training application and ephemeral CI/reproduction labs. The acknowledgement was confirmed before Day 4 implementation resumed. Signed material, API credentials, raw scanner matches, private paths and the complete SBOM remain outside public artifacts.

The current deferred register contains 180 underlying issues, including 0 critical and 91 high. Deferral is an explicit study decision, not remediation. Later work should prioritize compatible dependency fixes, tested traffic isolation, genuine-secret response if ever confirmed and human-reviewed behavior-preserving changes.

GenAI disclosure: the Codex coding assistant authored implementation, tests, documentation and bounded proposals. OpenAI GPT-6 Luna is the configured external triage model; unsuccessful live attempts and mock results are separated. Human review and remaining assessment dependencies are not invented.

| Reference | Location |
|---|---|
| Project evidence | docs/evidence-index.md and docs/project-completion.md |
| ZAP baseline behavior | https://www.zaproxy.org/docs/docker/baseline-scan/ |
| Kubernetes controls | https://kubernetes.io/docs/tasks/configure-pod-container/security-context/ |
| CIS benchmark scope | https://www.cisecurity.org/benchmark/kubernetes |
| OWASP SAMM v2 | https://owaspsamm.org/model/ |
| OpenAI model / price | https://developers.openai.com/api/docs/models/gpt-6-luna |
| OpenAI data controls | https://developers.openai.com/api/docs/guides/your-data |

Submission limits: successful external live triage, genuine PR review and hosted verification of the new critical-remediation revision remain open requirements. No grade, exhaustive detection, full CIS compliance or production readiness is asserted.